FLUCTUATION BOUNDS FOR ROTATIONALLY INVARIANT FIRST-PASSAGE PERCOLATION

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ABSTRACT. We prove that in rotationally invariant first-passage percolation on $\mathbb{R}^d$ with $d \geq 2$, the passage time T_n over a distance n satisfies $\mathbb{E}[\exp(|T_n - \mu n|/n^{\frac{1}{2}(1-\varepsilon)})] \leq C$ for some $\varepsilon > 0$, where μ is the time constant. This implies that its variance is at most a constant times $n^{1-\varepsilon}$, the bound that Basu, Sidoravicius and Sly [BSS23, BSS26] proved in the planar case. As in their work, our argument is an induction on the scale, but in place of their proof that the geodesic is chaotic, we use the averaging argument of Benjamini, Kalai and Schramm [BKS03] together with a logarithmic improvement of the Efron–Stein inequality due to Benaïm and Rossignol [BR08].

1. INTRODUCTION

In continuum first-passage percolation, one measures distances in $\mathbb{R}^d$ by integrating a random density along curves. A natural example starts from Gaussian white noise ω on $\mathbb{R}^d$ with $d \geq 2$. Convolving it with a smooth radial function χ supported in the closed unit ball, and applying a continuous function ψ with values between two positive constants, gives the random density

$$\rho := \psi(\chi * \omega)\,.$$

Thus the value of ρ at a point depends only on the noise within unit distance of that point, and ρ is equivariant under rigid motions. The *passage time* between two points $u, v \in \mathbb{R}^d$ is

$$T(u,v) := \inf_{\pi : u \to v} \int_\pi \rho(x)\,|dx|\,, \tag{1.1}$$

where the infimum is over rectifiable curves π from u to v. By subadditivity, the *time constant* $\mu := \lim_{t\to\infty} t^{-1}\mathbb{E}[T(0,te_1)]$ exists, and it is positive and finite since ρ lies between two positive constants.

Theorem 1.1. *For the density $\rho = \psi(\chi*\omega)$, there exist $\varepsilon > 0$ and $C < \infty$ such that, for every $x \in \mathbb{R}^d$ with $|x| \geq 1$,*

$$\mathbb{E}\Big[\exp\Big(\frac{|T(0,x) - \mu|x||}{C|x|^{\frac{1}{2}(1-\varepsilon)}}\Big)\Big] \leq 2\,.$$

Theorem 1.1 implies that $\mathrm{Var}[T(0,x)] \leq 2C^2|x|^{1-\varepsilon}$ for $|x| \geq 1$. It also improves the classical concentration on the scale $|x|^{1/2}$, due on the lattice to Kesten [Kes93, (1.15)] and, with Gaussian tails, to Talagrand [Tal95, (8.3.2)]. In first-passage percolation on $\mathbb{Z}^d$, with independent and identically distributed edge weights, Kesten proved that $\mathrm{Var}[T(0,x)] \leq C|x|$ under mild conditions, and asked for a bound by $|x|^{1-\varepsilon}$ [Kes93, Remark 1]. Benjamini, Kalai and Schramm [BKS03] improved his bound by a factor $1/\log|x|$ for weights taking two values with equal probability, and this has since been extended to broad classes of weights [BR08, DHS15]. Kesten's question remains open [ADH17, Question 13]. However, in a companion paper [AB26], we answer it positively in every direction in which the time constant satisfies a nondegenerate $C^{2,\alpha}$ condition. No such condition is known for any distribution of the weights; however, for exponential weights on $\mathbb{Z}^2$, strict convexity and differentiability of the limit shape were recently announced [Ope26].

For $d = 2$, with $\chi \geq 0$ and ψ smooth and increasing, this variance bound was proved by Basu, Sidoravicius and Sly [BSS23, BSS26]. Their proof is an induction on scales. Write $T_n := T(0, ne_1)$.

Since T_{Kn}, for a large integer K, can be compared with a sum of K independent passage times across slabs of width n, the Efron–Stein inequality gives $\mathrm{Var}[T_{Kn}] \le CK\,\mathrm{Var}[T_n]$. Replacing C by $1/2$ and iterating over the scales K^j gives a power saving. To show that C can be replaced by $1/2$, Basu, Sidoravicius and Sly prove that the geodesic is chaotic, in the sense of Chatterjee [Cha08]: after the noise is resampled in a small random fraction of space, the new geodesic has a small overlap with the old one. This carries out a suggestion of Kesten, who noted that a power might be gained by bounding the probability that an edge lies on the geodesic. Their proof uses the FKG inequality, which holds under their assumptions $\chi \ge 0$ and ψ increasing. It also uses estimates from [BSS23], which rely on planarity to bound the variance from below.

We follow a different multiscale strategy: we induct not on the variance but on an exponential moment of $|T_n - \mu n|$ on an explicit scale $n^{\frac{1}{2}(1-\varepsilon)}$. Exponential tails give bounds that hold simultaneously for many pairs of points, and the induction controls the mean as well, so we need neither the estimates of [BSS23] nor a lower bound on the variance. We also obtain the improvement on the Efron–Stein inequality differently. As Benjamini, Kalai and Schramm did on the lattice, we average the passage time over translations of its endpoints. Rotational invariance makes the limit shape a ball, so the fluctuation bounds of the induction keep near-geodesics close to the segment between their endpoints. Hence each piece of the noise affects only a small fraction of the translated passage times, and the inequality of Benaïm and Rossignol [BR08] gives a gain logarithmic in K. Our proof uses neither planarity nor the FKG inequality, so it applies in every dimension and to every χ and ψ as above.

1.1. General densities.

Assumption A. *Let $d \ge 2$, and let ω be Gaussian white noise or a homogeneous Poisson process on $\mathbb{R}^d$. The density $\rho(\omega, x)$ is jointly Borel measurable, and:*

1. Locality: $\rho(\omega, x) = \rho(\omega', x)$ *whenever ω and ω' agree on the closed unit ball centered at x;*
2. Boundedness: $0 < \inf_{\omega,x} \rho(\omega, x) \le \sup_{\omega,x} \rho(\omega, x) < \infty$;
3. Equivariance under rigid motions: $\rho(\omega \circ g^{-1}, gx) = \rho(\omega, x)$ *for every composition g of a rotation and a translation.*

The passage time T and the time constant μ are defined as above.

Theorem 1.2. *Under Assumption A, there exist $\varepsilon > 0$ and $C < \infty$ such that, for every $x \in \mathbb{R}^d$ with $|x| \ge 1$,*

$$\mathbb{E}\Big[\exp\Big(\frac{|T(0,x) - \mu|x||}{C|x|^{\frac{1}{2}(1-\varepsilon)}}\Big)\Big] \le 2\,.$$

Since $\psi(\chi * \omega)$ satisfies Assumption A also when ω is a Poisson process, Theorem 1.2 contains Theorem 1.1 and the Poisson case of the theorem of Basu, Sidoravicius and Sly.

1.2. Proof overview.

By stationarity and rotational invariance, $T(u, v)$ has the law of T_t with $t = |v - u|$. Replacing ρ by ρ/μ, we assume that $\mu = 1$, so that $\mathbb{E}[T_t] \ge t$ for every $t > 0$, by subadditivity. Fix a large integer K and a large constant A, and let $Q_s := As^{1/2 - 1/K}$. We induct on the following statement, for $s \ge 1$:

$$\mathbb{E}\big[\exp\big(|T_s - s|/Q_s\big)\big] \le 2\,. \tag{1.2}$$

It controls both the *random fluctuations* $T_s - \mathbb{E}[T_s]$ and the *nonrandom fluctuations* $\mathbb{E}[T_s] - s$. For small s, it holds if A is large, since ρ is bounded; this is the base case of the induction.

The induction step derives (1.2) at a distance Kn from (1.2) at the distances $s \le Kn/2$. By union bounds, the latter give an exponential bound at the distance Kn, but with a large constant:

$$\mathbb{E}\big[\exp\big(8|T_{Kn} - \mathbb{E}[T_{Kn}]|/Q_{Kn}\big)\big] \le B\,, \tag{a}$$

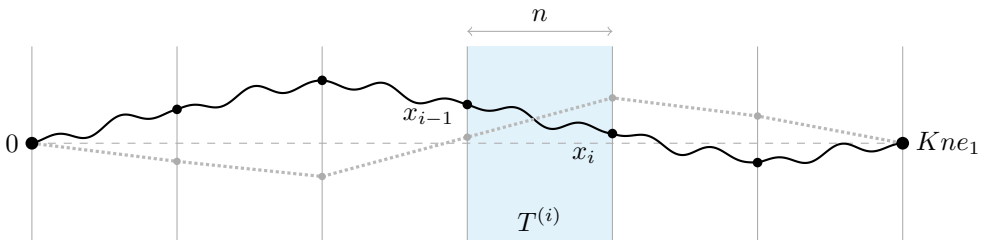


FIGURE 1. Slabs of width n between 0 and Kne_1: the black dots form a minimizing chain, joined by near-geodesics for the slab passage times $T^{(i)}$ (solid), and the grey dots joined by the dotted line form another chain.

where B does not depend on A or K. This bound cannot be iterated, since the constant would grow at each step. Our main estimate improves on the bound $\mathrm{Var}[T_{Kn}] \le CKQ_n^2 = CK^{2/K}Q_{Kn}^2$ given by the Efron–Stein inequality:

$$\mathrm{Var}[T_{Kn}] \le \delta_K Q_{Kn}^2\,, \qquad \text{with } \delta_K \to 0 \text{ as } K \to \infty. \tag{b}$$

We interpolate between (a) and (b) with the Cauchy–Schwarz inequality:

$$\mathbb{E}\big[\exp\big(4|T_{Kn} - \mathbb{E}[T_{Kn}]|/Q_{Kn}\big)\big] \le 1 + 4(B\delta_K)^{1/2}\,.$$

By subadditivity, $\mathbb{E}[T_{Kn}] \ge Kn$, and a separate argument gives

$$\mathbb{E}[T_{Kn}] - Kn \le Q_{Kn}/25\,. \tag{c}$$

For K large, the last two bounds give $\mathbb{E}[\exp(2|T_{Kn} - Kn|/Q_{Kn})] \le 2$, which is (1.2) at $s = Kn$ with $Q_{Kn}/2$ in place of Q_{Kn}.

To prove (a), (b) and (c), we compare T_{Kn} with independent passage times across the K slabs $[(i-1)n, in] \times \mathbb{R}^{d-1}$ (Figure 1). Here a *slab* of width s is the image of $[0,s] \times \mathbb{R}^{d-1}$ under a rigid motion, and its *faces* are the images of $\{0\} \times \mathbb{R}^{d-1}$ and $\{s\} \times \mathbb{R}^{d-1}$. For a closed set $P \subseteq \mathbb{R}^d$, let ρ_P be ρ at distance more than one from the boundary of P and $\sup \rho$ elsewhere in P, and, for $x, y \in P$, let

$$T_P(x,y) := \inf_\pi \int_\pi \rho_P(z)\,|dz|\,, \tag{1.3}$$

where the infimum is over the curves π from x to y in P. We write $T^{(i)}$ for T_P when P is the ith slab. By the locality in Assumption A, T_P is determined by $\omega|_P$, so the $T^{(i)}$ are independent, and, by (1.2), each fluctuates on the scale Q_n.

A *near-geodesic* from x to y in P is a rectifiable curve π from x to y in P with $\int_\pi \rho_P(z)\,|dz|$ at most $T_P(x,y) + 1$. We write $\Pi_{T_P}(x,y)$ for the set of these curves, and omit "in P" when $P = \mathbb{R}^d$. Modifying the near-geodesics from 0 to Kne_1 near the faces shows that

$$F := \min \sum_{i=1}^K T^{(i)}(x_{i-1}, x_i) \qquad \text{satisfies} \qquad \mathbb{E}\Big[\exp\Big(\frac{|F - T_{Kn}|}{CK\log^2(2+Kn)}\Big)\Big] \le 2\,, \tag{1.4}$$

where the minimum is over the *chains* $0 = x_0, x_1, \dots, x_K = Kne_1$ with $x_i \cdot e_1 = in$ and $|x_i - x_{i-1} - ne_1| \le n/4$. The error $K\log^2(2+Kn)$ is much smaller than Q_{Kn}. A *minimizing chain* is one that attains the minimum.

Step 1 (Section 4). We prove (a) and the following uniform version of (1.2). Fix $s \in [n, Kn]$ and two cubes of side $(sQ_s)^{1/2}$ in the faces $\{x \cdot e_1 = 0\}$ and $\{x \cdot e_1 = s\}$, whose centers differ by se_1 plus a vector of length at most s orthogonal to e_1 (Figure 2). Then

$$\mathbb{E}\Big[\exp\Big(\sup_{u,v} \big|T(u,v) - |v-u|\big|/(CQ_s)\Big)\Big] \le 2\,,$$

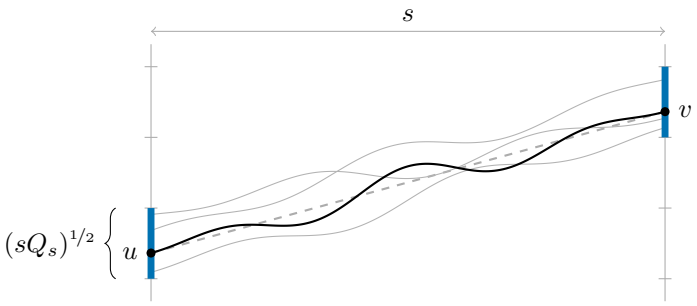


FIGURE 2. One slab of width s, its faces cut by tick marks into cubes of side $(sQ_s)^{1/2}$: the thick blue segments are two of these cubes, the thick black curve is a near-geodesic from u to v with the dashed segment $[u, v]$, and the grey curves are near-geodesics between other points of the two cubes.

where u and v range over the two cubes and $C = C(d, \rho)$. We need uniformity because the points of a minimizing chain are random. The side $(sQ_s)^{1/2}$ is the natural scale for moves orthogonal to e_1: moving an endpoint of $[0, se_1]$ by $\lambda(sQ_s)^{1/2}$ orthogonally to e_1 lengthens the segment by about $\lambda^2 Q_s/2$. To prove the bound for $s \le Kn/4$, we combine (1.2) at dyadic scales by a chaining argument. For larger s, we cut the segment of length s into a fixed number of slabs and take a union bound over the sequences of cubes visited by a chain. This also gives (a). The same union bound shows that $|x_i - x_{i-1} - ne_1|$ is at most of order $(KnQ_n)^{1/2}$ for a minimizing chain of F.

Step 2 (Section 5). We prove that near-geodesics stay close to the segment between their endpoints: for two cubes as in Step 1 with $s = n$, and S the slab between their faces,

$$\mathbb{E}\Big[\exp\Big(\sup_{u,v,\pi,x} \operatorname{dist}(x, [u, v])^2/(CnQ_n)\Big)\Big] \le 2\,, \tag{1.5}$$

where u and v range over the two cubes, π over the near-geodesics from u to v in S, and x over the points of π. We also prove that, for every $\eta > 0$ and some $h > 0$ depending on η,

$$\mathbb{E}\Big[\sup\big|T_S(u, v) - |v - u| - T_S(\widetilde{u}, \widetilde{v}) + |\widetilde{v} - \widetilde{u}|\big|^2\Big]^{1/2} \le \eta Q_n\,,$$

where the supremum is over the points $u, \widetilde{u}$ of one cube and $v, \widetilde{v}$ of the other with $|u - \widetilde{u}|$ and $|v - \widetilde{v}|$ at most $h(nQ_n)^{1/2}$. For (1.5), a curve that passes at distance $\lambda(nQ_n)^{1/2}$ from the segment is longer by order $\lambda^2 Q_n$, more than Step 1 allows when λ is large. To prove the continuity in the endpoints, we reroute the near-geodesics near their endpoints.

With $n^{1/2}$ in place of Q_n, transverse bounds like (1.5) go back to Newman and Piza [NP95, Theorem 6], in directions in which the limit shape of a lattice model is curved, and to Howard and Newman [HN01, Theorem 2.4] for Euclidean models, where isotropy makes the limit shape a ball; both obtain transverse fluctuations of order $n^{3/4}$, up to a factor n^{ε}. Alexander [Ale24, Theorems 1.3 and 1.6] proved uniform versions for isotropic random-graph models with exponential tails at all distances. In (1.5), the scale Q_n comes instead from (1.2) at smaller distances.

Step 3 (Section 6). We prove (b) with the following modified Poincaré inequality.

Proposition 1.3 (Benaïm and Rossignol [BR08, Proposition 2.1]). *Let $\alpha \in (0, 1]$, and let $X_1, X_2, \dots$ be independent random variables, each taking finitely many values, each with probability at least α. Then every square-integrable function f of $(X_j)_j$ satisfies*

$$\operatorname{Var}[f] \log \frac{\operatorname{Var}[f]}{\sum_j \mathbb{E}\big[|f - \mathbb{E}_j[f]|\big]^2} \le \frac{4}{\alpha} \sum_j \mathbb{E}\big[(f - \mathbb{E}_j[f])^2\big]\,,$$

where $\mathbb{E}_j[f]$ denotes the expectation of f with respect to X_j alone.

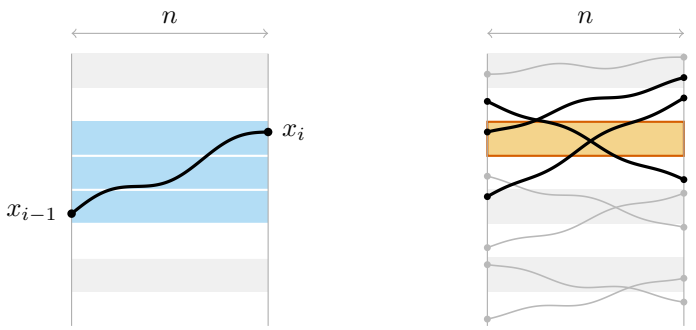


FIGURE 3. One slab of width n cut into blocks (grey and white bands): on the left, a near-geodesic from x_{i-1} to x_i crosses it and meets the three blue blocks; on the right, near-geodesics of $\lceil K^{2/3} \rceil = 8$ translates of F, with 0 and Kne_1 shifted by multiples of the block side, cross it, and the three black ones meet the orange block.

Benaïm and Rossignol assume the logarithmic Sobolev inequality $\mathrm{Ent}[g^2] \le (4/\alpha)\,\mathrm{Var}[g]$ for the functions g of each X_j, where $\mathrm{Ent}[g^2] := \mathbb{E}[g^2 \log g^2] - \mathbb{E}[g^2] \log \mathbb{E}[g^2]$. It holds here, since $g^2 \le \mathbb{E}[g^2]/\alpha$ gives $\mathrm{Ent}[g^2] \le \mathrm{Var}[g^2]/\mathbb{E}[g^2] \le 4\,\mathrm{Var}[g]/\alpha$.

Let L be a large integer, and let G range over the *blocks*

$$[(i-1)n, in] \times (nQ_n)^{1/2}\big(z + [0,L)^{d-1}\big)\,, \qquad 1 \le i \le K,\ z \in L\mathbb{Z}^{d-1}\,.$$

We apply Proposition 1.3 with variables X_G that approximate the normalized passage times $(T^{(i)}(u,v) - n)/Q_n$ across G. Each X_G is a function of $\omega|_G$ with finitely many values, each of probability at least a constant $\alpha_L > 0$ independent of n, and $\mathbb{E}_G$ is the expectation with respect to X_G alone. By Step 2, changing $\omega|_G$ changes F only if, before or after the change, a segment $[x_{i-1}, x_i]$ of a minimizing chain passes within distance of order $(nQ_n)^{1/2}$ of G. Proposition 1.3 is useful only if $\sum_G \mathbb{E}[|f - \mathbb{E}_G[f]|]^2$ is much smaller than $\sum_G \mathbb{E}[(f - \mathbb{E}_G[f])^2]$. We cannot prove this for F: the blocks containing 0 and Kne_1 always matter, and elsewhere we would need a lower bound on transverse fluctuations.

Following the averaging trick of Benjamini, Kalai and Schramm [BKS03, Lemma 3], we therefore apply Proposition 1.3 to the average of F over $\lceil K^{2/3} \rceil$ translates, whose slabs and endpoints are shifted by $mL(nQ_n)^{1/2}e_2$ for $0 \le m < \lceil K^{2/3} \rceil$ (see also [BR08, Lemma 5.6] and [DHS15, Section 3.1]). By Steps 1 and 2, for each block G, the near-geodesics joining the points of a minimizing chain of F meet on average at most order $K^{1/2}$ of the blocks $G + mL(nQ_n)^{1/2}e_2$ with $m \in \mathbb{Z}$ (Figure 3, left), so, by stationarity, each block is met by the near-geodesics of at most order $K^{1/2}$ of the translates on average (Figure 3, right). Let $\overline{F}$ be this average and $f := \mathbb{E}[\overline{F} \mid (X_G)_G]$; by Steps 1 and 2, the conditional variance $\mathbb{E}[\mathrm{Var}[\overline{F} \mid (X_G)_G]]$ is small. Resampling $\omega|_G$ changes f only through the translates whose near-geodesics meet G, a fraction of order $K^{1/2}/K^{2/3} = K^{-1/6}$ of them, so the two sums in Proposition 1.3 satisfy

$$\sum_G \mathbb{E}\big[(f - \mathbb{E}_G[f])^2\big] \le CKQ_n^2 \qquad \text{and} \qquad \sum_G \mathbb{E}\big[|f - \mathbb{E}_G[f]|\big]^2 \le CK^{-1/6}KQ_n^2\,.$$

Proposition 1.3 then gives $\mathrm{Var}[f] \le CKQ_n^2/(\alpha \log K)$.

Since the shifts are much smaller than $(KnQ_{Kn})^{1/2}$, the continuity in the endpoints of Step 2, at the distance Kn, and (1.4) show that $\overline{F}$ differs from T_{Kn} by a small multiple of Q_{Kn} in mean square, which gives (b).

Step 4 (Section 7). We prove (c) by showing that $\mathbb{E}[T_s] - s \le Q_s/100$ for $s = Kn/4$, which gives (c) by subadditivity. We place $\lceil s \rceil$ slabs of width s in a row, let F be the minimum (1.4) for

this row, and bound $\mathbb{E}[F] - s\lceil s\rceil$ from both sides:

$$\lceil s\rceil\Big(\frac{\mathbb{E}[T_s]-s}{1+\sup\rho} - C\kappa Q_s\Big) \le \mathbb{E}[F] - s\lceil s\rceil \le C\lceil s\rceil \log^2(2+s)\,.$$

For the lower bound, isotropy and subadditivity give $\mathbb{E}[T^{(i)}(u,v)] \ge s + (\mathbb{E}[T_s]-s)/(1+\sup\rho)$ for all points u and v on the faces of the ith slab. By Steps 1–3, $\mathbb{E}[\sup_{u,v}|T^{(i)}(u,v) - \mathbb{E}[T^{(i)}(u,v)]|] \le \kappa Q_s$, with κ small, where u and v range over two cubes of side $(sQ_s)^{1/2}$ in the faces of the ith slab. By Step 1, a minimizing chain moves by order $(sQ_s)^{1/2}$ per slab on average, so along it the fluctuations add up to at most $C\kappa\lceil s\rceil Q_s$ in mean. The upper bound follows from (1.4) for this row and the bound $\mathbb{E}[T_t] - t \le C(t\log(2+t))^{1/2}$ of Howard and Newman [HN01, Lemmas 4.1 and 4.2] at $t = s\lceil s\rceil$. Since $\log^2(2+s)$ is much smaller than Q_s, the two bounds force $\mathbb{E}[T_s] - s \le Q_s/100$ when κ is small.

In isotropic random-graph models with exponential tails at all distances, Alexander [Ale24, Theorem 1.5] bounds the nonrandom fluctuations by a large constant times the fluctuation scale. The induction needs the much smaller constant $1/100$, but only at one distance, and the small variance of Step 3 provides it.

1.3. **Notation and conventions.** We use the notation of [AKM19, (0.41)–(0.42)] for exponential bounds and its properties from [AKM19, Appendix A]. For $s, \theta > 0$ and a random variable X, we write

$$X \le \mathcal{O}_s(\theta) \quad\Longleftrightarrow\quad \mathbb{E}\big[\exp\big((X_+/\theta)^s\big)\big] \le 2\,, \qquad X_+ := \max\{X,0\}\,,$$
$$X \le Y + \mathcal{O}_s(\theta) \text{ and } Y \ge X - \mathcal{O}_s(\theta) \quad\Longleftrightarrow\quad X - Y \le \mathcal{O}_s(\theta)\,.$$

If $X \le \mathcal{O}_s(\theta)$, then $\mathbb{P}[X \ge \theta x] \le 2e^{-x^s}$ for $x \ge 0$ and $\mathbb{E}[X_+^p] \le C(p,s)\theta^p$ for $p \ge 1$. If $X_j \le \mathcal{O}_s(\theta)$ for $j \le J$, then

$$\max_{j\le J} X_j \le \mathcal{O}_s\big(C(s)\theta \log^{1/s}(2+J)\big)\,. \tag{1.6}$$

If $s \ge 1$ and $0 \le X_j \le \mathcal{O}_s(\theta_j)$ for every j, then

$$\sum_j X_j \le \mathcal{O}_s\Big(\sum_j \theta_j\Big)\,. \tag{1.7}$$

All constants and thresholds may depend on d and the law of the density, and we indicate only further dependence, as in $C(L)$. The constants $c, C > 0$ may change from line to line. We write $[u,v]$ for the segment from u to v, and $\angle(x,y) \in [0,\pi]$ for the angle between nonzero vectors x and y. We write $e_1,\dots,e_d$ for the standard basis of $\mathbb{R}^d$ and $\mathbf{1}_E$ for the indicator of an event E.

1.4. **Formalization.** All the results of this paper are formalized in the Lean 4 proof assistant, with the Mathlib library, in the repository

https://github.com/nitromannitol/FPP-variance

It also contains the companion paper [AB26] on lattice first-passage percolation, with the formalization of its main theorem, and checked lists of the errors and gaps found in [BSS23] and [BSS26].

1.5. **Use of artificial intelligence.** This paper grew out of a project, begun on October 21, 2026, to formalize the two papers of Basu, Sidoravicius and Sly [BSS23, BSS26] in the Lean 4 proof assistant with the help of AI agents. Checking their arguments in the detail that a formalization requires, the agents found a number of points that needed correction. Many have local corrections, but we did not obtain a corrected proof. While searching for another route, we found a much simpler argument, which is the one presented here and applies in every dimension. Its induction on scales comes from the work of Basu, Sidoravicius and Sly, and the present paper would not exist without their pioneering work.

An eight-page proof for a planar model was obtained by GPT-6 Astra agents, and its extension to every dimension by a GPT-6.1 Sol agent. An initial draft of the present paper was written by

Claude Opus 5.5 under our direction, and the authors subsequently heavily revised and rewrote this draft. The formalization described in Section 1.4 was written with the help of AI agents.

Acknowledgments. S. A. acknowledges support from the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme, grant agreement number 101200828.

2. Preliminaries

This section collects classical bounds on passage times and near-geodesics, the comparison of slab passage times with T, and a bound for sums of independent arrays along a sequence. Throughout, ρ satisfies Assumption A and, as in Section 1.2, $\mu = 1$.

Lemma 2.1. *For every $t \geq 1$,*

$$|T_t - \mathbb{E}[T_t]| \leq \mathcal{O}_2(Ct^{1/2}) \qquad \text{and} \qquad 0 \leq \mathbb{E}[T_t] - t \leq C(t\log(2+t))^{1/2}\,. \tag{2.1}$$

Proof. Both tails of the first bound follow from the inequalities of Boucheron, Lugosi and Massart [BLM03, Corollary 3 and Theorem 7], since resampling ω in one unit cube changes T_t by at most C, and the squares of the increases of T_t sum over the cubes to at most Ct. The second bound follows from the first as in Howard and Newman [HN01, Lemmas 4.1 and 4.2]. □

Lemma 2.1 implies that, away from its endpoints, a near-geodesic lies in narrow cones about the segment between them, uniformly over the endpoints in a ball.

Lemma 2.2. *Let $r \geq 1$. There exists a random variable W with $\log(2+r) \leq W \leq \mathcal{O}_2(C\log(2+r))$ such that, for all points a and b in the ball of radius r around 0, every point x of a near-geodesic from a to b with $|x-a| \geq CW^2$ satisfies $\angle(x-a, b-a) \leq 1/10$.*

Proof. We use Lemma 2.1 to make the straightness argument of Howard and Newman [HN01, Theorem 2.6 and Lemma 2.7] uniform in the endpoints. The near-geodesics between points of the ball lie in the ball of radius Cr around 0. Let $W \geq \log(2+r)$ be the smallest number such that $|T(x,y) - |y-x|| \leq C(W|y-x|^{1/2} + W^2)$ for all x and y in this larger ball. By Lemma 2.1, isotropy, the union bound (1.6) over the pairs of integer points in it, and $|T(x,y) - T(x',y')| \leq C(|x-x'| + |y-y'|)$, we have $W \leq \mathcal{O}_2(C\log(2+r))$. With this bound in place of their fluctuation bound, their argument proves the lemma. □

The proof uses the minimum (1.4) for several numbers of slabs. For a positive integer M and $s > 2$, let F be the minimum (1.4) over the chains across the M slabs $[(i-1)s, is] \times \mathbb{R}^{d-1}$, with $\rho^{(i)}$ and $T^{(i)}$ the ρ_P and T_P of the ith slab. For $\tau \in \mathbb{R}^d$, let F_τ be defined by (1.4) with the slabs, the endpoints and the points x_i translated by τ. The next lemma compares slab passage times with T, and proves the comparison of F with T_{Ms} stated in Section 1.2.

Lemma 2.3. *Let P be a slab of width $s > 2$, and let $z \in \mathbb{R}^d$. Then*

$$0 \leq \sup_{x,y}\big(T_P(x,y) - T(x,y)\big) \leq \mathcal{O}_1(C\log^2(2+s))\,, \tag{2.2}$$

where x and y range over the points on different faces of P with $|x-z| \leq s$ and $|y-x| \leq 5^{1/2}\,s$. Moreover,

$$0 \leq F - T_{Ms} \leq \mathcal{O}_1(CM\log^2(2+Ms))\,, \tag{2.3}$$

and the same bound holds for $F_\tau - T(\tau, \tau + Mse_1)$, for every $\tau \in \mathbb{R}^d$.

Proof. By stationarity, we may assume that $z = 0$ and $\tau = 0$. The lower bounds follow from $T_P \geq T$, which holds since $\rho_P \geq \rho$ on P, and from the triangle inequality. Let W be as in Lemma 2.2, with $r := 4s$ for (2.2) and $r := CMs$ for (2.3), so that $W^2 \leq \mathcal{O}_1(C\log^2(2+r))$.

For (2.2), suppose first that $s \geq CW^2$. Since $|y-x| \leq 5^{1/2}\, s$, the segment $[x,y]$ makes an angle at most $\arctan 2$ with the normal of P. Hence, by Lemma 2.2 from x and from y, the points of a near-geodesic from x to y at distance at least CW^2 from x and y lie at distance more than one from ∂P. Replacing its two end pieces by segments gives $T_P(x,y) \leq T(x,y) + CW^2$. If $s < CW^2$, the segment $[x,y]$ gives the same bound. This proves (2.2).

For (2.3), suppose first that $s \geq CW^2$. Let x_i be the first point of a near-geodesic from 0 to Mse_1 on the face $\{x \cdot e_1 = is\}$, for $0 < i < M$, and let $x_0 := 0$ and $x_M := Mse_1$. Since subarcs of near-geodesics are near-geodesics, Lemma 2.2, applied to the subarc from x_{i-1} to Mse_1 and to the reversed curve, gives $\angle(x_i - x_{i-1}, e_1) \leq 1/5$. Since $\tan 1/5 < 1/4$, the x_i are admissible in (1.4). The bound $T_P \leq T + CW^2$ in each slab then gives $F \leq T_{Ms} + 1 + CMW^2$. If $s < CW^2$, the points ise_1 give $F \leq CMW^2$. This proves (2.3). □

Moment bounds for fixed indices do not directly bound a sum along a random sequence of indices. The next lemma allows random sequences, at the cost of their squared increments.

Lemma 2.4. *Let $M \geq 1$ be an integer, and let $Z_1, \ldots, Z_M$ be independent random arrays of nonnegative random variables $Z_i(j,k)$, indexed by $j,k \in \mathbb{Z}^{d-1}$, with $\mathbb{E}[Z_i(j,k)^{4d}] \leq 1$ for all i,j,k. Then, with $k_0 := 0$ and a constant C depending only on d,*

$$\mathbb{E}\Big[\sup_{k_1,\ldots,k_M \in \mathbb{Z}^{d-1}} \Big\{ \sum_{i=1}^M Z_i(k_{i-1},k_i) - \sum_{i=1}^M |k_i - k_{i-1}|^2 \Big\}\Big] \leq CM\,.$$

Proof. For $k = (k_1, \ldots, k_M)$ in $(\mathbb{Z}^{d-1})^M$, with $k_0 = 0$, let

$$S(k) := \sum_{i=1}^M Z_i(k_{i-1},k_i) \qquad \text{and} \qquad P(k) := \sum_{i=1}^M |k_i - k_{i-1}|^2\,,$$

so that the lemma bounds $\mathbb{E}[\sup_k (S(k) - P(k))]$. Let $V_r := \max\{S(k) : P(k) \leq 2^r M\}$ for each integer $r \geq 0$. If $2^{r-1}M < P(k) \leq 2^r M$, then $S(k) - P(k) \leq V_r - 2^{r-1}M$. Hence

$$\mathbb{E}\Big[\sup_k \big(S(k) - P(k)\big)\Big] \leq \mathbb{E}[V_0] + \sum_{r \geq 1} \mathbb{E}\big[(V_r - 2^{r-1}M)_+\big] \leq \mathbb{E}[V_0] + \sum_{r \geq 1} \frac{\mathbb{E}[V_r^2]}{2^{r-1}M}\,,$$

and it suffices to show that $\mathbb{E}[V_r^2]^{1/2} \leq C2^{r/4}M$ for every r.

Fix r. For each k, we write $Z_{(1)}(k) \geq \cdots \geq Z_{(M)}(k)$ for the entries $Z_i(k_{i-1},k_i)$ in decreasing order. For $1 \leq a \leq M$, let $Y_a := \max\{Z_{(a)}(k) : P(k) \leq 2^r M\}$, so that $V_r \leq \sum_a Y_a$. For $\lambda > 0$, if $Y_a \geq \lambda$, then $Z_{(a)}(k) \geq \lambda$ for some k with $P(k) \leq 2^r M$, so a of the entries $Z_i(k_{i-1},k_i)$ are at least λ. These entries occupy a triples (i, k_{i-1}, k_i), which are determined by their indices i and by $2a$ displacements in $\mathbb{Z}^{d-1}$ of total length at most $\sum_i |k_i - k_{i-1}| \leq (MP(k))^{1/2} \leq 2^{r/2}M$. Hence the set of these a triples takes at most $(eM/a)^a (C2^{r/2}M/a)^{2a(d-1)}$ values. For each such set, the a entries at its triples lie in distinct arrays, so, by independence and Markov's inequality for their $4d$th moments, they are all at least λ with probability at most λ^{-4da}. A union bound over these sets therefore gives

$$\mathbb{P}[Y_a \geq \lambda] \leq \Big(\frac{eM}{a}\Big)^a \Big(\frac{C2^{r/2}M}{a}\Big)^{2a(d-1)} \lambda^{-4da} \leq \Big(\frac{C2^{r/4}(M/a)^{1/2}}{\lambda}\Big)^{4da}\,,$$

so $\mathbb{E}[Y_a^2]^{1/2} \leq C2^{r/4}(M/a)^{1/2}$, and summing over a by Minkowski's inequality gives $\mathbb{E}[V_r^2]^{1/2} \leq C2^{r/4}M$. □

3. The induction

We prove Theorem 1.2 by induction on the length. For an integer $K \geq 10$ and $A \geq 1$, let $Q_s := As^{1/2-1/K}$ for $s > 0$, as in Section 1.2. For a length $N \geq 2$, the *induction hypothesis* is

$$|T_s - s| \leq \mathcal{O}_1(Q_s) \qquad \text{for } 1 \leq s \leq N/2\,, \tag{3.1}$$

which is (1.2) for these s, and the conclusion of the induction step is

$$|T_N - N| \leq \mathcal{O}_1(Q_N/2)\,. \tag{3.2}$$

Proposition 3.1 (Induction step)**.** *There exist an integer $K \geq 10$ and $N_0 \geq 2$ such that (3.1) implies (3.2) for every $A \geq 1$ and every $N \geq N_0$.*

Proof of Theorem 1.2, given Proposition 3.1. This is immediate by Proposition 3.1 and induction on the length, with $\varepsilon = 2/K$ and μA as the constant of Theorem 1.2, since $|T_s - s| \leq Cs$ gives $|T_s - s| \leq \mathcal{O}_1(Q_s)$ for $s \in [1, N_0]$ once A is large, depending on K. □

The rest of the paper proves Proposition 3.1. By (2.1), $|T_N - N| \leq \mathcal{O}_1(C(N\log(2+N))^{1/2})$, which gives (3.2) if $Q_N \geq 2C(N\log(2+N))^{1/2}$. We may therefore assume that, with this constant C,

$$Q_N \leq 2C(N\log(2+N))^{1/2}\,. \tag{3.3}$$

From now on, we assume (3.1) and (3.3).

4. Passage times between cubes

In this section we extend the induction hypothesis (3.1) in two ways, for arbitrary numbers $0 < \beta_1 \leq \beta_2$: to bounds uniform over the pairs of points in two cubes on opposite faces of a slab of width $s \in [\beta_1 N, N]$, and to lengths $t \in [\beta_1 N, \beta_2 N]$. For $s > 2$, a rigid motion g, numbers $\sigma, \sigma' \in [0, 4(sQ_s)^{1/2}]$ and $y \in \mathbb{R}^{d-1}$, let

$$\begin{gathered} I := g\big(\{0\} \times [-\sigma/2, \sigma/2]^{d-1}\big)\,, \qquad I' := g\big(\{s\} \times (y + [-\sigma'/2, \sigma'/2]^{d-1})\big)\,, \\ S := g\big([0,s] \times \mathbb{R}^{d-1}\big)\,. \end{gathered} \tag{4.1}$$

We first extend (3.1) from points to boxes by chaining over dyadic subboxes. A *box* is the image under a rigid motion of a product of d compact intervals, some of which may be single points.

Lemma 4.1. *For $16 \leq t \leq N/2$ and every box B centered at te_1, of diameter at most $t/8$, contained in the $16(d\,tQ_t)^{1/2}$-neighborhood of the line $\mathbb{R}e_1$,*

$$\sup_{z \in B}\{T(0,z) - z \cdot e_1\} \leq \mathcal{O}_1(CQ_t)\,. \tag{4.2}$$

By stationarity and isotropy, the same holds for $T(x,z) - (z-x)\cdot e$ over boxes centered at $x + te$ contained in the $16(d\,tQ_t)^{1/2}$-neighborhood of $x + \mathbb{R}e$, for every $x \in \mathbb{R}^d$ and unit vector e.

Proof. For an integer $j \geq 0$, the *dyadic subboxes* of B of generation j are the 2^{dj} boxes obtained by cutting each edge of B into 2^j intervals of equal length. For such a subbox D with center c, let $p(D) := c - 2^{-j}te_1$, so that $p(B) = 0$. Let $\mathcal{A}_j$ be the set of the $2^{d(j+1)}$ pairs $(p(D), p(D'))$, where D has generation j and $D' \subseteq D$ has generation $j+1$. For $(y,w) \in \mathcal{A}_j$, the vector $w - y$ is 2^{-j-1} times the vector from 0 to a vertex of B, so

$$\frac{15}{32}2^{-j}t \leq |w-y| \leq 2^{-j}t \qquad \text{and} \qquad |w-y| - (w-y)\cdot e_1 \leq C2^{-j}Q_t\,. \tag{4.3}$$

Let j_0 be the largest integer with $2^{-j_0}t \geq 4$. For $j < j_0$, the pairs in $\mathcal{A}_j$ satisfy $1 \leq |w-y| \leq N/2$. Hence (3.1), isotropy, $Q_{2^{-j}t} \leq 2^{-2j/5}Q_t$ and the union bound (1.6) show that the maximum X_j of $(T(y,w) - |w-y|)_+$ over $(y,w) \in \mathcal{A}_j$ satisfies $X_j \leq \mathcal{O}_1(C(1+j)2^{-2j/5}Q_t)$.

Let $z \in B$, let $D_0 = B \supseteq D_1 \supseteq \cdots$ be nested subboxes containing z, with D_j of generation j, and let $y_j := p(D_j)$. Since $y_0 = 0$ and $|z - y_{j_0}| \le 9$, the triangle inequality and (4.3) give

$$T(0,z) - z \cdot e_1 \le \sum_{j<j_0} \big(T(y_j, y_{j+1}) - (y_{j+1} - y_j) \cdot e_1 \big) + C \le \sum_{j<j_0} \big(X_j + C 2^{-j} Q_t \big) + C \,.$$

The right side does not depend on z, and, by (1.7), it is at most $\mathcal{O}_1(CQ_t)$. □

We prove the next proposition with Lemma 4.1 and the triangle inequality through points of $\mathbb{R}e_1$, first for $s \le N/4$, where (3.1) applies at all the lengths involved, and then for $s > N/4$, after bounding T_t for every $t \in [\beta_1 N, \beta_2 N]$.

Proposition 4.2 (Passage times between cubes)**.** *If N is sufficiently large, depending only on K and β_1, then, for every $s \in [\beta_1 N, N]$ and all cubes I, I' as in (4.1) with $|y| \le s$,*

$$\sup_{u \in I,\, v \in I'} |T_P(u,v) - |v-u|| \le \mathcal{O}_1(CQ_s) \,, \tag{4.4}$$

both for $P = \mathbb{R}^d$ and for $P = S$.

Proof of Proposition 4.2 for $s \le N/4$. By stationarity and isotropy, we may assume that the centers of I and I' are 0 and re_1, with $s \le r \le 2^{1/2} s$. By the triangle inequality, for $u \in I$ and $v \in I'$,

$$\begin{aligned} T(u,v) &\ge T\big(-s/16 e_1, (r + s/16) e_1\big) - T\big(-s/16 e_1, u\big) - T\big(v, (r+s/16)e_1\big) \,, \\ T(u,v) &\le T\big(u, s/16 e_1\big) + T\big(s/16 e_1, (r - s/16) e_1\big) + T\big((r - s/16)e_1, v\big) \,. \end{aligned}$$

Since $r + s/8 < 2s \le N/2$, the induction hypothesis (3.1) gives $|T(x,z) - |z-x|| \le \mathcal{O}_1(CQ_s)$ for the two terms $T(x,z)$ of the display with $x, z \in \mathbb{R}e_1$. Since $Q_{s/16} \ge Q_s/4$, Lemma 4.1 with $t = s/16$ and $e = \pm e_1$ gives $T(x,z) \le (z-x)\cdot e_1 + \mathcal{O}_1(CQ_s)$ for the four terms of the display that involve u or v, uniformly in u and v, for N large. In each line of the display, the terms $(z-x)\cdot e_1$ and $|z-x|$ of these bounds add up to $(v-u)\cdot e_1$, so, by (1.7), $|T(u,v) - (v-u)\cdot e_1| \le \mathcal{O}_1(CQ_s)$ uniformly in u and v. Since $0 \le |v-u| - (v-u)\cdot e_1 \le CQ_s$, this proves (4.4) for $P = \mathbb{R}^d$. For N large, $|v-u| \le 5^{1/2}\, s$, so (2.2) with $P = S$ and $z = 0$ adds $\mathcal{O}_1(C \log^2(2+N)) \le \mathcal{O}_1(Q_s)$. □

We now cut the faces of the slabs of (1.4) into cubes of side $(sQ_s)^{1/2}$ (Figure 2). For M, s and F as in Section 2, let

$$I_i(j) := \{is\} \times (sQ_s)^{1/2} \big(j + [0,1]^{d-1} \big) \qquad \text{for } 0 \le i \le M \text{ and } j \in \mathbb{Z}^{d-1} \,, \tag{4.5}$$

and let $x_0, \dots, x_M$ be the lexicographically least chain attaining the minimum in (1.4). Translating ω and the endpoints by a vector orthogonal to e_1 translates the x_i by the same vector. For an array $Z_i(j,k)$ indexed by $1 \le i \le M$ and $j, k \in \mathbb{Z}^{d-1}$, we write $Z_i(u,v) := Z_i(j,k)$ for the lexicographically greatest j and k with $u \in I_{i-1}(j)$ and $v \in I_i(k)$.

The next lemma bounds F and the transverse increments of its minimizing chain, by a union bound over the sequences $(k_i)_i$ for which a chain visits the cubes $I_i(k_i)$.

Lemma 4.3. *If N is sufficiently large, depending only on K and β_1, then, for every positive integer M and every $s \in [\beta_1 N, N/4]$,*

$$|F - Ms| + \frac{1}{s} \sum_{i=1}^{M} |x_i - x_{i-1} - s e_1|^2 \le CMQ_s + \mathcal{O}_1(CQ_s) \,. \tag{4.6}$$

Proof. For $1 \le i \le M$ and $j, k \in \mathbb{Z}^{d-1}$ with $(sQ_s)^{1/2} |k - j| \le s/2$, let $u \in I_{i-1}(j)$ and $v \in I_i(k)$, so that these cubes are as in (4.1) with $\sigma = \sigma' = (sQ_s)^{1/2}$, with S the ith slab and with $|y| \le s/2$. For N large, $|v-u| \ge s + cQ_s |k-j|^2 - CQ_s$. Hence Proposition 4.2, with $P = S$ the ith slab, gives

$$Z_i(j,k) := \Big(s + cQ_s |k-j|^2 - \inf_{I_{i-1}(j) \times I_i(k)} T^{(i)} \Big)_+ \le \mathcal{O}_1(C_1 Q_s) \,.$$

The arrays Z_i are independent over i, so, for $k_0 = 0, k_1, \dots, k_M$ with $(sQ_s)^{1/2}|k_i - k_{i-1}| \le s/2$,

$$\mathbb{E}\Big[\exp\Big(\sum_{i=1}^{M} \frac{Z_i(k_{i-1},k_i) - \frac{c}{2}Q_s|k_i-k_{i-1}|^2}{C_1Q_s}\Big)\Big] \le \prod_{i=1}^{M} 2e^{-c|k_i-k_{i-1}|^2/(2C_1)}\,.$$

The sum of these products over $k_1, \dots, k_M \in \mathbb{Z}^{d-1}$ is at most C^M. Hence a union bound gives

$$\sup_{k_1,\dots,k_M}\Big\{Ms + \frac{c}{2}Q_s\sum_{i=1}^{M}|k_i-k_{i-1}|^2 - \sum_{i=1}^{M}\inf_{I_{i-1}(k_{i-1})\times I_i(k_i)} T^{(i)}\Big\} \le CMQ_s + \mathcal{O}_1(CQ_s)\,.$$

The same argument, applied to the single chain $x_i = ise_1$, gives $F \le Ms + CMQ_s + \mathcal{O}_1(CQ_s)$. Let k_i be the lexicographically greatest index with $x_i \in I_i(k_i)$; since $|x_i - x_{i-1} - se_1| \le s/4$, we have $(sQ_s)^{1/2}|k_i - k_{i-1}| \le s/2$ for N large. Then

$$F \ge \sum_{i=1}^{M}\inf_{I_{i-1}(k_{i-1})\times I_i(k_i)} T^{(i)} \qquad \text{and} \qquad |x_i - x_{i-1} - se_1|^2 \le 2sQ_s\big(|k_i-k_{i-1}|^2 + d - 1\big)\,,$$

so the bound on the supremum gives (4.6). □

For s and N as in Lemma 4.3, let $Z_1, \dots, Z_M$ be independent random arrays of nonnegative random variables with $\mathbb{E}[(Z_i(j,k) - |j-k|^2)_+^{4d}] \le 1$ for all i, j, k. With k_i the lexicographically greatest index with $x_i \in I_i(k_i)$, (4.6) gives $\mathbb{E}[\sum_i |k_i - k_{i-1}|^2] \le CM$, so Lemma 2.4, applied to the arrays $(Z_i(j,k) - |j-k|^2)_+$, gives

$$\mathbb{E}\Big[\sum_{i=1}^{M} Z_i(x_{i-1}, x_i)\Big] \le CM\,. \tag{4.7}$$

The next lemma gives the bound (a) of Section 1.2, from Lemma 4.3 and (2.3) with a fixed number of slabs.

Lemma 4.4. *If N is sufficiently large, depending only on K, β_1, β_2, then, for every $t \in [\beta_1 N, \beta_2 N]$,*

$$|T_t - t| \le C(\beta_2)Q_t + \mathcal{O}_1(Q_t/100)\,. \tag{4.8}$$

Proof. Fix an integer M, depending only on β_2, such that $s := t/M \le N/4$ and $C_1M^{-2/5} \le 1/200$, where C_1 is the constant of the term $\mathcal{O}_1(CQ_s)$ in (4.6). Since $s \ge \beta_1 N/M$, Lemma 4.3 with β_1/M in place of β_1, and (2.3), give $|T_t - t| \le CMQ_s + \mathcal{O}_1(C_1Q_s + CM\log^2(2+N))$. Since $Q_s \le M^{-2/5}Q_t$ and $CM\log^2(2+N) \le Q_t/200$ for N large, this proves (4.8). □

Proof of Proposition 4.2 for $N/4 < s \le N$. The proof for $s \le N/4$ applies, with (4.8) in place of (3.1) at the lengths $r \pm s/8 \in [N/8, 2N]$. □

5. Transverse fluctuations and continuity in the endpoints

We fix a length $n \in [\beta_1 N, N]$, and cubes I, I' and a slab S as in (4.1) with $s = n$ and $|y| \le n/2$. We bound the distance from the curves in $\Pi_{T_S}(u,v)$, for $u \in I$ and $v \in I'$, to $[u,v]$, and the modulus of continuity of $T_S(u,v) - |v-u|$ in u and v.

To bound the distance, we extend the curves to two fixed points a and b on the line through the centers of the cubes. Since the limit shape is a ball, the results of Section 4 show that an extended curve through a point at distance δ from this line has passage time at least $|b-a| + c\delta^2/n - \mathcal{O}_1(CQ_n)$, uniformly in the curve.

Lemma 5.1 (Transverse fluctuations)**.** *If N is sufficiently large, depending only on K and β_1, then*

$$(nQ_n)^{-1/2}\sup_{u\in I, v\in I'}\sup_{\pi\in\Pi_{T_S}(u,v)}\sup_{x\in\pi}\operatorname{dist}(x,[u,v]) \le \mathcal{O}_2(C)\,. \tag{5.1}$$

Since ρ is bounded, moving the endpoints by $h(nQ_n)^{1/2}$ changes $T_S(u,v)$ by at most $Ch(nQ_n)^{1/2}$, which is much larger than Q_n. The next lemma shows that $T_S(u,v) - |v-u|$ changes only by a small multiple of Q_n; its proof reroutes a near-geodesic near its endpoints, where fluctuation bounds at shorter distances apply.

Lemma 5.2 (Continuity in the endpoints)**.** *For every $\eta > 0$, there exists $h > 0$, depending only on η, such that, if N is sufficiently large, depending only on K, η, β_1, then*

$$\mathbb{E}\Big[\sup_{\substack{u,\widetilde{u}\in I,\ v,\widetilde{v}\in I' \\ |u-\widetilde{u}|,\,|v-\widetilde{v}|\le h(nQ_n)^{1/2}}} \big|T_S(u,v) - T_S(\widetilde{u},\widetilde{v}) - |v-u| + |\widetilde{v}-\widetilde{u}|\big|^{8d} \Big]^{1/8d} \le \eta Q_n \,. \tag{5.2}$$

Proof of Lemma 5.1. By stationarity and isotropy, we may assume that the centers of I and I' are 0 and re_1, with $n \le r \le 2n$. Write $z = (z_1, w)$ with $z_1 := e_1 \cdot z$ and $w \in \mathbb{R}^{d-1}$. The points of I and I' lie within distance $2(d\,nQ_n)^{1/2}$ of 0 and re_1, and the curves in $\Pi_{T_S}(u,v)$ have length at most $C(|v-u|+1)$. Hence, for N large, the curves in (5.1) lie in

$$\Lambda := \{(z_1, w) : -C_0 n \le z_1 \le r + C_0 n,\ |w| \le C_0 n\}$$

for a constant C_0. Let $a := -2C_0 n e_1$ and $b := (r + 2C_0 n)e_1$. For a rectifiable curve π, let $T(\pi) := \int_\pi \rho(z)\,|dz|$. We compare an upper bound on $T(a,u) + T(\pi) + T(v,b)$ (the black curves in Figure 4) with a lower bound on $T(a,z) + T(z,b)$ for $z \in \Lambda$ (the orange dashed curve).

We first prove the upper bound

$$\sup_{u\in I,\, v\in I'} \ \sup_{\pi\in\Pi_{T_S}(u,v)} \big\{T(a,u) + T(\pi) + T(v,b)\big\} \le |b-a| + \mathcal{O}_1(CQ_n)\,.$$

Since $\rho \le \rho_S$, we have $T(\pi) \le T_S(u,v) + 1 \le |v-u| + \mathcal{O}_1(CQ_n)$, by (4.4). By the triangle inequality through the points $-n/16 e_1$ and $(r + n/16)e_1$, Lemma 4.1 at the length $n/16$ and (4.8) at the length $(2C_0 - 1/16)n$ give $T(a,u) \le e_1 \cdot (u-a) + \mathcal{O}_1(CQ_n)$ and $T(v,b) \le e_1 \cdot (b-v) + \mathcal{O}_1(CQ_n)$, uniformly in u and v. Since $|v-u| - e_1 \cdot (v-u) \le CQ_n$, the upper bound follows by (1.7).

We next prove the lower bound: for some $c > 0$,

$$\sup_{z=(z_1,w)\in\Lambda} \Big\{ |b-a| + \frac{c|w|^2}{n} - T(a,z) - T(z,b) \Big\} \le \mathcal{O}_1(CQ_n)\,. \tag{5.3}$$

We cover Λ by boxes and bound $T(a,z)$ and $T(z,b)$ from below through points at distance $n/16$ from the center of each box. For integers $j \ge 0$ and $k \in \mathbb{Z}^{d-1}$, let $t_j := -C_0 n + jn/500$ and

$$B_{j,k} := [t_j - n/1000, t_j + n/1000] \times (nQ_n)^{1/2}\big(k + [-1,1]^{d-1}\big)\,;$$

we use only the boxes that meet Λ, which cover it. Let $x_\pm := (t_j \pm n/16, (nQ_n)^{1/2}k)$. For N large, Lemma 4.1 at the length $n/16$ gives $T(z, x_+) \le e_1 \cdot (x_+ - z) + \mathcal{O}_1(CQ_n)$ for $z \in B_{j,k}$. By (4.8) at

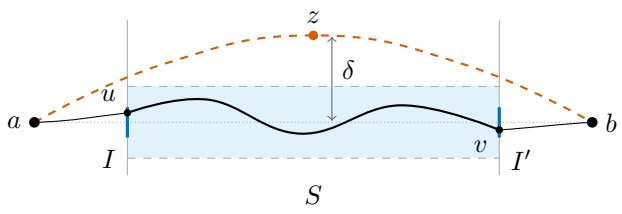


FIGURE 4. The slab S with the cubes I and I' (thick blue) on its faces, a curve in $\Pi_{T_S}(u,v)$ (black) extended to a and b (thin black), and a curve from a to b (orange dashed) through a point z at distance δ from the line through a and b (dotted), outside the shaded band of width $C(nQ_n)^{1/2}$.

the length $|x_+ - a| \in [C_0 n, Cn]$, isotropy and the triangle inequality, for $z \in B_{j,k}$,

$$T(a,z) \geq e_1 \cdot (z-a) + \frac{nQ_n|k|^2}{|x_+ - a| + e_1 \cdot (x_+ - a)} - \mathcal{O}_1(CQ_n) \geq e_1 \cdot (z-a) + cQ_n|k|^2 - \mathcal{O}_1(CQ_n)\,.$$

With

$$X_{j,k} \coloneqq \sup_{z \in B_{j,k}} \bigl(|b-a| + cQ_n|k|^2 - T(a,z) - T(z,b)\bigr)_+\,,$$

this bound and the same bound for $T(z,b)$, using x_-, give $X_{j,k} \leq \mathcal{O}_1(CQ_n)$. Moreover, $|w|^2/n \leq Q_n(2|k|^2 + C)$ for $z = (z_1, w) \in B_{j,k}$, so

$$|b-a| + \frac{c|w|^2}{4n} - T(a,z) - T(z,b) \leq X_{j,k} - \frac{c}{2} Q_n |k|^2 + CQ_n\,.$$

Summing the tails of $X_{j,k} - cQ_n|k|^2/2$ over the boxes proves (5.3).

Finally, let $Y \coloneqq \sup\{|w| : (z_1, w) \in \pi,\ \pi \in \Pi_{T_S}(u,v),\ u \in I,\ v \in I'\}$. If z lies on $\pi \in \Pi_{T_S}(u,v)$, then $T(a,z) + T(z,b) \leq T(a,u) + T(\pi) + T(v,b)$, so the two bounds give $Y^2 \leq \mathcal{O}_1(CnQ_n)$, that is, $Y \leq \mathcal{O}_2(C(nQ_n)^{1/2})$. Since $|y| \leq n/2$, the normal of S makes an angle at most $\arctan 1/2$ with e_1, so the points $z = (z_1, w)$ of S satisfy $-|w|/2 \leq z_1 \leq r + |w|/2$. Hence the points z of the curves satisfy $\operatorname{dist}(z, [u,v]) \leq 2Y + 2(d\,nQ_n)^{1/2}$. □

Proof of Lemma 5.2. We may again assume that the centers of I and I' are 0 and re_1. Fix $h \in (0, 4^{-3})$, let $\gamma \coloneqq h^{1/3}$, let Z be the supremum of $|T_S(u,v) - T_S(\widetilde{u},\widetilde{v}) - |v-u| + |\widetilde{v} - \widetilde{u}||$ over the points in (5.2), and let N be large, depending on K, h, β_1. We bound Z separately on the event E that the left side of (5.1) is at least $1/\gamma$, and off E. Then $\mathbb{P}[E] \leq 2e^{-c\gamma^{-2}}$ by (5.1), and $\mathbb{E}[Z^{16d}]^{1/16d} \leq CQ_n$ by (4.4). Hence the Cauchy–Schwarz inequality gives $\mathbb{E}[Z^{8d}\mathbf{1}_E]^{1/8d} \leq Ce^{-c\gamma^{-2}}Q_n$.

Off E, we reroute the curves near their endpoints. With g as in (4.1), let $S_1 \coloneqq g([0, \gamma n] \times \mathbb{R}^{d-1})$ and $S_2 \coloneqq g([(1-\gamma)n, n] \times \mathbb{R}^{d-1})$, and let $H_1 \coloneqq g(\{\gamma n\} \times \mathbb{R}^{d-1})$ and $H_2 \coloneqq g(\{(1-\gamma)n\} \times \mathbb{R}^{d-1})$. Let $u, \widetilde{u} \in I$ and $v, \widetilde{v} \in I'$ with $|u - \widetilde{u}|, |v - \widetilde{v}| \leq h(nQ_n)^{1/2}$, let $\pi \in \Pi_{T_S}(u,v)$, let x_1 be its first point in H_1, and let x_2 be its last point in H_2. Since $\rho \leq \rho_S \leq \rho_{S_1}$ on S_1, and similarly on S_2, replacing the pieces of π before x_1 and after x_2 by curves in $\Pi_{T_{S_1}}(\widetilde{u}, x_1)$ and $\Pi_{T_{S_2}}(x_2, \widetilde{v})$ gives

$$T_S(\widetilde{u},\widetilde{v}) - T_S(u,v) \leq T_{S_1}(\widetilde{u}, x_1) - T(u, x_1) + T_{S_2}(x_2, \widetilde{v}) - T(x_2, v) + 3\,.$$

Let

$$D \coloneqq \sup\bigl|T_P(z,z') - |z'-z|\bigr|\,,$$

where the supremum is over $P \in \{\mathbb{R}^d, S_1\}$, $z \in I$ and $z' \in H_1$, and over $P \in \{\mathbb{R}^d, S_2\}$, $z \in I'$ and $z' \in H_2$, with $\operatorname{dist}(z', [0, re_1]) \leq (\gamma^{-1} + 2d^{1/2})(nQ_n)^{1/2}$. Off E, the points x_1 and x_2 satisfy this condition. These pairs of points lie in at most $C\gamma^{-4d}$ pairs of cubes as in (4.1) with $s = \gamma n$ and $|y| \leq \gamma n$, and with S_1 or S_2 in place of S. Hence (4.4) and the union bound (1.6) give $D \leq \mathcal{O}_1(CQ_{\gamma n}\log(2/\gamma))$. Writing $|x| = e_1 \cdot x + \phi(x)$, with $\phi(x) \coloneqq |x| - e_1 \cdot x$, the terms $e_1 \cdot x$ cancel, and

$$\begin{aligned} &T_S(\widetilde{u},\widetilde{v}) - |\widetilde{v} - \widetilde{u}| - T_S(u,v) + |v-u| \\ &\qquad \leq 4D + 3 + \phi(x_1 - \widetilde{u}) - \phi(x_1 - u) + \phi(\widetilde{v} - x_2) - \phi(v - x_2) - \phi(\widetilde{v} - \widetilde{u}) + \phi(v-u)\,. \end{aligned}$$

If $e_1 \cdot x \geq 0$, then $|\nabla\phi(x)| \leq 2|x - (e_1 \cdot x)e_1|/|x|$. This is at most $C(1+\gamma^{-1})(nQ_n)^{1/2}/(\gamma n)$ on the segments joining the arguments of the first two pairs of ϕ terms, and at most $C(Q_n/n)^{1/2}$ on the segment joining those of the last pair. The arguments in each pair differ by at most $2\gamma^3(nQ_n)^{1/2}$, so the ϕ terms add up to at most $C\gamma Q_n$. Exchanging the roles of (u,v) and $(\widetilde{u},\widetilde{v})$ gives $Z \leq 4D + 3 + C\gamma Q_n$ off E. Since $1 \leq \gamma Q_n$ and $Q_{\gamma n} \leq \gamma^{2/5} Q_n$,

$$\mathbb{E}[Z^{8d}]^{1/8d} \leq C\bigl(e^{-c\gamma^{-2}} + \gamma^{2/5}\log(2/\gamma) + \gamma\bigr)Q_n\,.$$

Since $\gamma = h^{1/3}$, we have $C(e^{-c\gamma^{-2}} + \gamma^{2/5}\log(2/\gamma) + \gamma) \leq \eta$ if h is small, depending only on η. □

6. A logarithmic gain of variance

In this section we prove the bound (b) of Section 1.2: $\mathrm{Var}[T_{Kn}] \le \delta_K Q_{Kn}^2$ for $Kn \in [N/8, N]$. We let δ_K and δ_L denote numbers depending only on K, respectively on L, that tend to zero as $K \to \infty$, respectively $L \to \infty$, and may change from line to line.

Let n be a length with $Kn \in [N/8, N]$, and let F, $T^{(i)}$, $I_i(j)$ and x_i be as defined before Lemma 4.3, with $M = K$ and $s = n$. For an integer $L \ge 1$ and vectors $a_i \in \{0, \dots, L-1\}^{d-1}$, we shift the blocks of Section 1.2 in the ith slab by $(nQ_n)^{1/2} a_i$, so that they are the sets $[(i-1)n, in] \times (nQ_n)^{1/2}(z + [0,L)^{d-1})$ for $z \in a_i + L\mathbb{Z}^{d-1}$.

For each block G, let σ_G be a random variable determined by $\omega|_G$. The *resampling* of G is the noise ω^G that coincides with ω outside G and whose restriction to G is drawn, given ω, from the conditional law of $\omega|_G$ given σ_G, independently for distinct blocks, so that (ω^G, ω) has the law of (ω, ω^G). With σ_G constant, $\omega^G|_G$ is an independent copy of $\omega|_G$. Let $T^{(i),G}$ and F^G be $T^{(i)}$ and F computed with ω^G.

Changing $\omega|_G$ can increase $T^{(i)}(u,v)$ only if some curve in $\Pi_{T^{(i)}}(u,v)$ passes within distance one of G. For $1 \le i \le K$ and $j, k \in \mathbb{Z}^{d-1}$ with $(nQ_n)^{1/2}|k-j| \le n/2$, we therefore let

$$R_i(j,k) := (d-1)^{1/2} + 2 + (nQ_n)^{-1/2} \sup_{u \in I_{i-1}(j),\, v \in I_i(k)} \sup_{\pi \in \Pi_{T^{(i)}}(u,v)} \sup_{x \in \pi} \operatorname{dist}(x, [\bar{u}, \bar{v}]) \,,$$

where $\bar{u}$ and $\bar{v}$ are the centers of $I_{i-1}(j)$ and $I_i(k)$, the *tube*

$$U_i(j,k) := \left\{ x : \operatorname{dist}(x, [\bar{u}, \bar{v}]) \le R_i(j,k)(nQ_n)^{1/2} \right\},$$

and

$$Y_i(j,k) := \frac{1}{Q_n^2} \sum_G \Big[\sup_{u \in I_{i-1}(j),\, v \in I_i(k)} \big(T^{(i),G}(u,v) - T^{(i)}(u,v) \big)_+ \Big]^2 ,$$

where the sum is over the blocks G of the ith slab. For the other j, k, we let $R_i(j,k) = Y_i(j,k) = 0$. For N large, $nQ_n \ge 1$, so $U_i(j,k)$ contains the points within distance one of the curves in $\Pi_{T^{(i)}}(u,v)$ for $u \in I_{i-1}(j)$ and $v \in I_i(k)$. The arrays $Y_1, \dots, Y_K$ are independent, and, by Assumption A(1), the term of G in $Y_i(j,k)$ vanishes unless G meets $U_i(j,k)$.

The next lemma bounds the cost of resampling blocks along the minimizing chain, and the widths of the tubes.

Lemma 6.1. *If N is sufficiently large, depending only on K, then*

$$\mathbb{E}\Big[\sum_{i=1}^K R_i(x_{i-1}, x_i)^2 \Big] \le CK \,, \tag{6.1}$$

and, for every integer $L \ge 1$, every choice of the vectors a_i and of the random variables σ_G, every $i \in \{1, \dots, K\}$ and every pair j, k with $(nQ_n)^{1/2}|k-j| \le n/2$,

$$\mathbb{E}[Y_i(j,k)^{16d}]^{1/16d} \le C(1 + |j-k|) \qquad \text{and} \qquad \sum_G \mathbb{E}[(F^G - F)^2] \le CKQ_n^2 \,, \tag{6.2}$$

where the sum is over the blocks of all K slabs.

Proof. Lemma 5.1 at the length $n \ge N/(8K)$ gives $R_i(j,k) \le \mathcal{O}_2(C)$. The arrays R_i^2 are independent over i, so (4.7), applied to a small multiple of R_i^2, gives (6.1).

For the first bound in (6.2), we cover the ith slab by the *cells*

$$[(i-1)n, in] \times (nQ_n)^{1/2} \big(z + [0,1)^{d-1} \big) \,, \qquad z \in \mathbb{Z}^{d-1} .$$

The term of a block G in $Y_i(j,k)$ is at most the number of cells of G that meet $U_i(j,k)$, times $Q_n^{-2} \sup_{u,v} |T^{(i),G}(u,v) - T^{(i)}(u,v)|^2$. Since ω^G has the law of ω, every moment of $Q_n^{-1} \sup_{u,v} |T^{(i),G}(u,v) - T^{(i)}(u,v)|$ is bounded by (4.4), uniformly in G, in L and in

the σ_G. The cell of z meets $U_i(j,k)$ only if $R_i(j,k) \geq \operatorname{dist}(z,[j,k]) - C$, where $[j,k] \subseteq \mathbb{R}^{d-1}$ is the segment between the indices. Hence Hölder's inequality and $R_i(j,k) \leq \mathcal{O}_2(C)$ bound the $16d$th moment of the term of the cell of z in this bound by $Ce^{-c\operatorname{dist}(z,[j,k])^2}$. Summing over z, by Minkowski's inequality, gives the bound.

For the second bound, evaluating F^G at the points x_i gives, for N large,

$$\sum_G (F^G - F)_+^2 \leq Q_n^2 \sum_{i=1}^K Y_i(x_{i-1}, x_i)\,.$$

By exchangeability, $(F - F^G)_+$ has the law of $(F^G - F)_+$, so

$$\sum_G \mathbb{E}\big[(F^G - F)^2\big] \leq 2Q_n^2\, \mathbb{E}\Big[\sum_{i=1}^K Y_i(x_{i-1}, x_i)\Big]\,. \tag{6.3}$$

For $|j-k| \geq 1$, the first bound in (6.2) gives

$$\mathbb{E}\big[(Y_i(j,k) - |j-k|^2)_+^{4d}\big]^{1/4d} \leq |j-k|^{-6}\mathbb{E}\big[Y_i(j,k)^{16d}\big]^{1/4d} \leq C|j-k|^{-2}\,.$$

Hence (4.7), applied to a small multiple of the arrays Y_i, gives $\mathbb{E}[\sum_i Y_i(x_{i-1}, x_i)] \leq CK$, and (6.3) gives the second bound. □

To apply Proposition 1.3, we approximate the normalized passage times $(T^{(i)}(u,v) - n)/Q_n$ across each block G by a function of $\omega|_G$ with finitely many values.

Lemma 6.2. *Let $L \geq 16$ be an integer. There exists $\alpha_L > 0$, depending only on L, such that the following holds if N is sufficiently large, depending only on K and L. For every choice of the vectors a_i and every $i \in \{1, \dots, K\}$, there exist random variables X_G, one for each block G of the ith slab, such that X_G is determined by $\omega|_G$, takes finitely many values, each a bounded function of (u,v) and each with probability at least α_L, and satisfies*

$$\mathbb{E}\Big[\sup_{u,v}\Big|\frac{T^{(i)}(u,v) - n}{Q_n} - X_G(u,v)\Big|^{8d}\Big]^{1/8d} \leq \delta_L\,, \tag{6.4}$$

where u and v range over the points of the two faces of G at distance at least $(LnQ_n)^{1/2}$ from the other blocks of the ith slab. Moreover, for every block G of the ith slab and every τ orthogonal to e_1 such that $G + \tau$ is a block, $X_{G+\tau}(u+\tau, v+\tau)$, computed with ω translated by τ, equals $X_G(u,v)$.

Proof. Step 1. We compare $T^{(i)}$ with T_G, which is defined by (1.3) for the closure of G and determined by $\omega|_G$. On G, we have $\rho_G \geq \rho^{(i)}$, with equality at distance more than one from the other blocks of the ith slab. Hence $T_G(u,v) \geq T^{(i)}(u,v)$ for $u, v \in G$, with equality if the curves in $\Pi_{T^{(i)}}(u,v)$ stay at distance more than one from the other blocks. Let E be the event that, for some u and v on the faces of G at distance at least $(LnQ_n)^{1/2}/2$ from the other blocks, some curve in $\Pi_{T^{(i)}}(u,v)$ comes within distance one of another block. By (5.1) for the at most $CL^{2(d-1)}$ pairs of cubes $I_{i-1}(j)$ and $I_i(k)$ covering these points, $\mathbb{P}[E] \leq CL^{2(d-1)}e^{-cL}$.

Step 2. We discretize T_G. Let $G = [(i-1)n, in] \times (nQ_n)^{1/2}(z + [0,L)^{d-1})$ and $\kappa \in (0, (d-1)^{-1/2}]$. For $u = ((i-1)n, (nQ_n)^{1/2}(z+w))$ and $v = (in, (nQ_n)^{1/2}(z+w'))$, let u_κ and v_κ be u and v with w and w' replaced by $\kappa\lfloor w/\kappa \rfloor$ and $\kappa\lfloor w'/\kappa \rfloor$, coordinatewise, and

$$\Psi_G(u,v) := \kappa\Big\lfloor \frac{1}{\kappa} \max\Big\{-L^4, \min\Big\{L^4, \frac{T_G(u_\kappa, v_\kappa) - n}{Q_n}\Big\}\Big\}\Big\rfloor\,.$$

Then Ψ_G is determined by $\omega|_G$. As (u_κ, v_κ) takes at most $\lceil L/\kappa \rceil^{2(d-1)}$ values, and $\Psi_G(u,v)$ lies in $\kappa\mathbb{Z} \cap [-L^4 - \kappa, L^4]$, the function Ψ_G takes at most $J := (2L^4/\kappa + 2)^{\lceil L/\kappa \rceil^{2(d-1)}}$ values. For u and v

as in (6.4), the points u_κ and v_κ are at distance at least $(LnQ_n)^{1/2}/2$ from the other blocks. Hence Step 1 gives

$$|T^{(i)}(u,v) - n - Q_n\Psi_G(u,v)| \le |T^{(i)}(u,v) - T^{(i)}(u_\kappa,v_\kappa)| + \big(|T^{(i)}(u_\kappa,v_\kappa) - n| - L^4Q_n\big)_+ + 2L^4Q_n\mathbf{1}_E + \kappa Q_n\,.$$

The cubes covering the faces of G, enlarged to side $4(nQ_n)^{1/2}$, form at most $CL^{2(d-1)}$ pairs as in (4.1) with $s = n$ and $|y| \le CL(nQ_n)^{1/2} \le n/2$. On these pairs, we have $\big||v-u|-n\big| \le CL^2Q_n$, and $|v-u|$ changes by at most $CL\kappa Q_n$ when u and v are replaced by u_κ and v_κ. Hence (5.2), applied to each of these pairs, gives $\mathbb{E}[\sup_{u,v}|T^{(i)}(u,v) - T^{(i)}(u_\kappa,v_\kappa)|^{8d}] \le (\delta_L Q_n)^{8d}$ if κ is small, depending on L. By (4.4) and (1.6), $\sup_{u,v}|T^{(i)}(u_\kappa,v_\kappa) - n| \le CL^2Q_n + \mathcal{O}_1(CQ_n\log(2+L))$, so

$$\mathbb{E}\Big[\sup_{u,v}\big(|T^{(i)}(u_\kappa,v_\kappa) - n| - L^4Q_n\big)_+^{8d}\Big] \le (\delta_L Q_n)^{8d}\,.$$

Also, $\mathbb{E}[(2L^4Q_n\mathbf{1}_E)^{8d}] = (2L^4Q_n)^{8d}\mathbb{P}[E] \le (\delta_L Q_n)^{8d}$. Hence (6.4) holds with Ψ_G in place of X_G.

Step 3. We merge the values of small probability. We write Ψ_G as a function of the coordinates (w,w') of Step 2. For every τ orthogonal to e_1 such that $G+\tau$ is a block, $\Psi_{G+\tau}$, computed with ω translated by τ, is then the same function as Ψ_G. Hence, by stationarity, $p(\psi) := \mathbb{P}[\Psi_G = \psi]$ does not depend on the block G of the ith slab. Fix ψ^* with $p(\psi^*) = \max p$, and let $X_G := \Psi_G$ if $p(\Psi_G) \ge \alpha_L$ and $X_G := \psi^*$ otherwise. The X_G have the translation property of the lemma. Since Ψ_G takes at most J values, $p(\psi^*) \ge 1/J$, so every value of X_G has probability at least α_L if $\alpha_L \le 1/J$. Since $|X_G - \Psi_G| \le 3L^4$, and $X_G = \Psi_G$ off the event $\{p(\Psi_G) < \alpha_L\}$, which has probability at most $\alpha_L J$, the replacement adds at most $3L^4(\alpha_L J)^{1/8d}$ to the left side of (6.4). This is at most $1/L$ if α_L is small, depending on L and κ, and κ depends only on L. □

From now on, the X_G, for the blocks G of all K slabs, are random variables as in Lemma 6.2. Since X_G is determined by $\omega|_G$ and distinct blocks have disjoint interiors, the X_G of distinct blocks are independent. We average over translations: for $0 \le m < \lceil K^{2/3}\rceil$, let F_m be F_τ of Section 2 with $\tau = mL(nQ_n)^{1/2}e_2$, let F_m^G be F_m computed with ω^G, and let $\overline{F} := \lceil K^{2/3}\rceil^{-1}\sum_m F_m$. With $f := \mathbb{E}[\overline{F} \mid (X_G)_G]$,

$$\mathrm{Var}[\overline{F}] = \mathbb{E}\big[\mathrm{Var}[\overline{F} \mid (X_G)_G]\big] + \mathrm{Var}[f]\,.$$

The next lemma bounds the first term. We bound the second by Proposition 1.3, with the X_G as the variables and $\mathbb{E}_G[f]$ the expectation of f with respect to X_G alone.

Lemma 6.3. *Let $L \ge 16$ be an integer, and let ω^G be the resampling of G with $\sigma_G = X_G$. If N is sufficiently large, depending only on K and L, then*

(6.5)
$$\frac{1}{L^{K(d-1)}}\sum_a\sum_G \mathbb{E}\big[(F^G - F)^2\big] \le \delta_L K Q_n^2 \qquad \text{and} \qquad \frac{1}{L^{K(d-1)}}\sum_a \mathbb{E}\big[\mathrm{Var}[\overline{F} \mid (X_G)_G]\big] \le \delta_L K Q_n^2\,,$$

where $a = (a_1,\dots,a_K)$ ranges over $(\{0,\dots,L-1\}^{d-1})^K$; the blocks, the X_G and ω^G depend on a.

Proof. We begin with the first bound in (6.5), with the a_i independent of the rest and included in probabilities and expectations. Let $Y_i(j,k)$ be defined with this resampling, and let $\eta \in (0,1]$. We show that $\mathbb{E}[(Y_i(j,k) - \eta|j-k|^2)_+^{4d}]^{1/4d} \le \eta$ for all i,j,k if L is large depending on η. Then (4.7), applied to the arrays Y_i/η, gives $\mathbb{E}[\sum_i Y_i(x_{i-1},x_i)] \le C\eta K$, and (6.3) gives the first bound in (6.5).

If $|j-k| \ge 1$, then (6.2) gives $\mathbb{E}[(Y_i(j,k) - \eta|j-k|^2)_+^{4d}]^{1/4d} \le C\eta^{-3}|j-k|^{-2}$, which is at most η if $|j-k|$ is large depending on η. It remains to show that $\mathbb{E}[Y_i(j,k)^{4d}]$ tends to zero as $L \to \infty$ for each fixed value of $|j-k|$, uniformly in i. Let G be the block containing the center $\bar{u}$ of $I_{i-1}(j)$. Let E be the event that $\bar{u}$ lies within distance $C(1+|j-k|+L^{1/2})(nQ_n)^{1/2}$ of another block or that $R_i(j,k) > L^{1/2}$. Since a_i is uniform, (5.1) gives $\mathbb{P}[E] \le C(1+|j-k|+L^{1/2})/L + Ce^{-cL}$. Off E,

only G contributes to $Y_i(j,k)$, and the cubes $I_{i-1}(j)$ and $I_i(k)$ lie in the region of (6.4). The noise ω^G has the law of ω, and X_G takes the same value at ω and ω^G, so, off E,

$$\frac{1}{Q_n}\sup_{u,v}\bigl(T^{(i),G}(u,v)-T^{(i)}(u,v)\bigr)_+ \\ \le \sup_{u,v}\Bigl|\frac{T^{(i),G}(u,v)-n}{Q_n}-X_G(u,v)\Bigr|+\sup_{u,v}\Bigl|\frac{T^{(i)}(u,v)-n}{Q_n}-X_G(u,v)\Bigr|,$$

and (6.4) bounds both terms. On E, we use Hölder's inequality and (6.2).

For the second bound, given $(X_G)_G$, the $\omega|_G$ are conditionally independent, and $\omega^G|_G$ is a conditionally independent copy of $\omega|_G$. The Efron–Stein inequality [BLM03, Proposition 1], extended to countably many independent variables by martingale convergence, applied conditionally to $\overline{F}$, with Jensen's inequality over m, gives

$$\mathbb{E}\bigl[\mathrm{Var}[\overline{F}\mid (X_G)_G]\bigr]\le\frac{1}{2\lceil K^{2/3}\rceil}\sum_{m,G}\mathbb{E}\bigl[(F_m^G-F_m)^2\bigr].$$

Translations by multiples of $L(nQ_n)^{1/2}e_2$ map blocks to blocks and, by Lemma 6.2, the X_G of a block to that of the translated block. Hence, by stationarity, the sum over G is the same for every m, and the first bound in (6.5) bounds the right side. □

The next lemma bounds the two sums in Proposition 1.3.

Lemma 6.4. *Let $L\ge 16$ be an integer. If N is sufficiently large, depending only on K and L, then, for every choice of the vectors a_i,*

$$\sum_G\mathbb{E}\bigl[(f-\mathbb{E}_G[f])^2\bigr]\le CKQ_n^2 \qquad \text{and} \qquad \sum_G\mathbb{E}\bigl[|f-\mathbb{E}_G[f]|\bigr]^2\le CK^{-1/6}KQ_n^2.$$

Proof. Let ω^G be the resampling of G with σ_G constant, so that

$$f-\mathbb{E}_G[f]=\frac{1}{\lceil K^{2/3}\rceil}\sum_m\mathbb{E}\bigl[F_m-F_m^G\bigm|(X_H)_H\bigr].$$

Let G lie in the ith slab, and let x^m and $\widetilde{x}^m$ be the lexicographically least chains attaining F_m and F_m^G. Evaluating F_m^G at x^m and F_m at $\widetilde{x}^m$ shows that $F_m^G=F_m$ unless G meets $U_i(x_{i-1}^m,x_i^m)$, or $U_i(\widetilde{x}_{i-1}^m,\widetilde{x}_i^m)$ computed with ω^G.

The tube $U_i(x_{i-1},x_i)$ meets at most

$$C\Bigl(1+\frac{|x_i-x_{i-1}-ne_1|}{(nQ_n)^{1/2}}+R_i(x_{i-1},x_i)\Bigr)$$

translates of G by multiples of $L(nQ_n)^{1/2}e_2$. By (4.6) and (6.1), the squares of these numbers, summed over i, have mean at most CK, so, by the Cauchy–Schwarz inequality, each has mean at most $CK^{1/2}$. The translation by $-mL(nQ_n)^{1/2}e_2$ maps F_m to F and G to such a translate. Hence, by stationarity, for an independent m uniform in $\{0,\dots,\lceil K^{2/3}\rceil-1\}$, the block G meets either tube of F_m with probability at most $2CK^{1/2}/\lceil K^{2/3}\rceil\le CK^{-1/6}$ (Figure 3). Jensen's inequality, and the Cauchy–Schwarz inequality for the joint law of m and ω, give

$$\mathbb{E}\bigl[(f-\mathbb{E}_G[f])^2\bigr]\le\sum_m\frac{\mathbb{E}[(F_m-F_m^G)^2]}{\lceil K^{2/3}\rceil} \qquad \text{and} \qquad \mathbb{E}\bigl[|f-\mathbb{E}_G[f]|\bigr]^2\le CK^{-1/6}\sum_m\frac{\mathbb{E}[(F_m-F_m^G)^2]}{\lceil K^{2/3}\rceil}.$$

By stationarity, (6.2) holds also for F_m, and summing over G proves both bounds. □

Proposition 6.5 (The gain)**.** *If K is sufficiently large and N is sufficiently large depending on K, then*

$$\mathrm{Var}[T_{Kn}]\le\delta_KQ_{Kn}^2 \qquad \text{for every } n \text{ with } Kn\in[N/8,N].$$

Proof. Let n satisfy $Kn \in [N/8, N]$, and let $L \geq 16$ be an integer, to be chosen below depending only on K. All the bounds below hold for N large depending on K and L, uniformly in n and A.

We first bound $\mathrm{Var}[f]$, for every choice of the vectors a_i. Proposition 1.3 with $\alpha = \alpha_L$, which applies by Lemma 6.2, and Lemma 6.4 give

$$\mathrm{Var}[f] \log\Big(\frac{\mathrm{Var}[f]}{CK^{-1/6}KQ_n^2}\Big) \leq \frac{CKQ_n^2}{\alpha_L}\,.$$

If $\mathrm{Var}[f] \geq K^{-1/12}KQ_n^2$, then the logarithm is at least $\log(K^{1/12}/C)$, which is at least $(\log K)/24$ for K large. Hence $\mathrm{Var}[f] \leq (K^{-1/12} + C/(\alpha_L \log K))KQ_n^2$.

Since $\overline{F}$ does not depend on the vectors a_i, we may average the decomposition of $\mathrm{Var}[\overline{F}]$ over them. Lemma 6.3 and the bound on $\mathrm{Var}[f]$ then give $\mathrm{Var}[\overline{F}] \leq (\delta_L + K^{-1/12} + C/(\alpha_L \log K))KQ_n^2$.

We now let L grow slowly with K. The error δ_L needs $L \to \infty$, the term $C/(\alpha_L \log K)$ needs $\alpha_L \log K \to \infty$, and the comparison of $\overline{F}$ with T_{Kn} below needs $L \leq K^{1/24}$. We choose L as the largest integer in $[16, K^{1/24}]$ with $\alpha_L \geq (\log K)^{-1/2}$. It tends to infinity with K, so $\mathrm{Var}[\overline{F}] \leq \delta_K KQ_n^2$.

It remains to compare $\overline{F}$ with T_{Kn}. For the slab $S := [0, Kn] \times \mathbb{R}^{d-1}$, we have $\rho^{(i)} \geq \rho_S$ on the ith slab, so concatenating curves gives $T(\tau, \tau + Kne_1) \leq T_S(\tau, \tau + Kne_1) \leq F_\tau$. Since $(nQ_n/(KnQ_{Kn}))^{1/2} = K^{-3/4+1/(2K)}$ and $L \leq K^{1/24}$, the vectors τ of the translates satisfy $|\tau| \leq \lceil K^{2/3} \rceil L(nQ_n)^{1/2} \leq 2K^{-1/24+1/(2K)}(KnQ_{Kn})^{1/2}$. Hence (5.2) at the length Kn with this slab S, the bound of Lemma 2.3 on $F_\tau - T(\tau, \tau + Kne_1)$, and Jensen's inequality give $\mathbb{E}[(\overline{F} - T_{Kn})^2]^{1/2} \leq \delta_K Q_{Kn}$. Since $KQ_n^2 = K^{2/K}Q_{Kn}^2$ and $K^{1/K} \leq 2$, these bounds give $\mathrm{Var}[T_{Kn}]^{1/2} \leq \delta_K Q_{Kn}$. □

7. The mean and the induction step

In this section we show that $\mathbb{E}[T_s] - s \leq Q_s/100$ when $\mathrm{Var}[T_r/Q_r]$ is small for $r \in [s, 2s]$, and we complete the proof of the induction step, Proposition 3.1.

Lemma 7.1 (Nonrandom fluctuations)**.** *If N is sufficiently large, depending only on K, then $\mathbb{E}[T_s] - s \leq Q_s/100$ for every $s \in [N/8, N/4]$ such that $\sup_{s \leq r \leq 2s} \mathrm{Var}[T_r/Q_r]$ is at most c.*

Proof. Let $F, T^{(i)}, I_i(j)$ and x_i be as defined before Lemma 4.3, for $M = \lceil s \rceil$ slabs of width s. For $1 \leq i \leq \lceil s \rceil$ and $j, k \in \mathbb{Z}^{d-1}$, let

$$X_i(j,k) := \frac{1}{Q_s} \sup_{u \in I_{i-1}(j),\, v \in I_i(k)} |T^{(i)}(u,v) - \mathbb{E}[T^{(i)}(u,v)]| \qquad \text{if } (sQ_s)^{1/2}|k-j| \leq s/2\,,$$

and $X_i(j,k) := 0$ otherwise. The arrays X_i are independent over i.

We first show that, for every $\kappa \in (0,1]$, we have $\mathbb{E}[X_i(j,k)^{4d}]^{1/4d} \leq \kappa$ if $\sup_{s \leq r \leq 2s} \mathrm{Var}[T_r/Q_r]$ is small and N is large, depending on κ. For $u \in I_{i-1}(j)$ and $v \in I_i(k)$ with $(sQ_s)^{1/2}|k-j| \leq s/2$, we have $s \leq |v-u| \leq 2s$ for N large, and $T(u,v)$ has the law of $T_{|v-u|}$. By the hypothesis on the variance, (2.2) and (4.4), $D := (T^{(i)}(u,v) - \mathbb{E}[T^{(i)}(u,v)])/Q_s$ satisfies $\mathbb{E}[D^{8d}] \leq C$ and $\mathbb{E}[D^2] \leq C \sup_{s \leq r \leq 2s} \mathrm{Var}[T_r/Q_r] + C\log^4(2+s)/Q_s^2$, so, by Hölder's inequality, $\mathbb{E}[D^{4d}] \leq C\mathbb{E}[D^2]^c$. For a fixed $h \in (0,1]$, the same holds, with C depending also on h, for the maximum of $|D|$ over a mesh of $I_{i-1}(j) \times I_i(k)$ of spacing $h(sQ_s)^{1/2}$. Subtracting $|v-u|$ from both $T^{(i)}(u,v)$ and its mean changes nothing, so (5.2) extends this to the cubes. Choosing h small, then the variance small and then N large, all depending on κ, proves the claim.

For $r \geq s$, we have $\mathbb{E}[T_r] \geq r$ and, by subadditivity, $\mathbb{E}[T_r] \geq \mathbb{E}[T_s] - (r-s)\sup\rho$. Averaging these two bounds with the weights $\sup\rho/(1+\sup\rho)$ and $1/(1+\sup\rho)$ gives $\mathbb{E}[T_r] \geq s + (\mathbb{E}[T_s] - s)/(1+\sup\rho)$. For u and v as above, since $T^{(i)} \geq T$, this bound at $r = |v-u|$ gives

$$T^{(i)}(u,v) - s \geq \frac{\mathbb{E}[T_s] - s}{1 + \sup\rho} - Q_s X_i(j,k)\,.$$

By (4.7), applied to the arrays X_i/κ, we have $\mathbb{E}[\sum_i X_i(x_{i-1}, x_i)] \le C\kappa\lceil s\rceil$. Summing the bound on $T^{(i)}(u,v) - s$ at $u = x_{i-1}$ and $v = x_i$ therefore gives

$$\mathbb{E}[F] - s\lceil s\rceil \ge \lceil s\rceil \Big(\frac{\mathbb{E}[T_s] - s}{1 + \sup \rho} - C\kappa Q_s \Big) .$$

By (2.3) and (2.1) at the length $s\lceil s\rceil \le 2s^2$, the left side is at most $C\lceil s\rceil \log^2(2+s)$. Hence $\mathbb{E}[T_s] - s \le C\kappa Q_s + C\log^2(2+s)$. Since $Q_s \ge s^{1/2-1/K}$, this is at most $Q_s/100$ if κ is small and N is large. □

Proof of Proposition 3.1. Let $t \in [N/8, N]$. By (4.8), $\mathbb{E}[\exp(8|T_t - \mathbb{E}[T_t]|/Q_t)] \le C$. Hence the Cauchy–Schwarz inequality gives

$$\mathbb{E}\big[\exp\big(4|T_t - \mathbb{E}[T_t]|/Q_t\big)\big] \le 1 + C\operatorname{Var}[T_t/Q_t]^{1/2} .$$

By Proposition 6.5 with $n = t/K$, we may fix K so that, for N large depending on K, $C\operatorname{Var}[T_t/Q_t]^{1/2} \le 1$ and $\operatorname{Var}[T_t/Q_t]$ is as small as Lemma 7.1 requires. With $t = N$, this gives $|T_N - \mathbb{E}[T_N]| \le \mathcal{O}_1(Q_N/4)$. Lemma 7.1 at $s = N/4$ and subadditivity give

$$0 \le \mathbb{E}[T_N] - N \le 4\big(\mathbb{E}[T_{N/4}] - N/4\big) \le Q_N/25 .$$

Hence (3.2) follows by (1.7), and all the thresholds on N depend only on K. □

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